Arrival · Aspire

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Pair

What does heterozygous Aa mean?

Help: Compare the two alleles.

2. Gamete

How many alleles for this gene go in one gamete?

Help: One parent contributes one to the offspring pair.

3. Options

Which gamete alleles can an Aa parent produce?

Help: Separate the pair, not copy Aa into every margin.

4. Combine

A gamete with A combines with a gamete with a. What is the offspring genotype? Give a reason.

Help: Use one allele from each.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W05-L2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Aspire

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Construct

Complete Aa × Aa with labelled gametes and four offspring boxes.

Help: Combine one letter from the top and one from the side.

2. Compare

Construct Aa × aa and state the chance of aa.

Help: The aa parent contributes only a.

3. Critique a square

A pupil places AA and aa in the margins of Aa × Aa. Explain and repair the error.

Help: The margins contain gametes, not diploid offspring.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit ticket · Aspire

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

1. Construct

For Aa × Aa, list the four predicted genotype boxes.

Help: Combine the row and column labels.

2. Critique a square

A pupil places AA and aa in the margins of Aa × Aa. Explain and repair the error.

Help: The margins contain gametes, not diploid offspring.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W05-L2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Build the cross, then check the chance

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Complete the four boxes for Aa × Aa. Use one allele from each margin. Then change the second parent to aa and compare the two crosses.

Cross	Parent 1 gametes	Parent 2 gametes
Aa × Aa	A or a	A or a
Aa × aa	A or a	a only

My evidence and reasoning

One limit or useful check

My response: _____

Keep your first answer. Hear the feedback. Choose one real change, or explain why to keep it. Ask for a read-back if it helps.

Made from the web lesson SCI-LAUNCH-A2-W05-L2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Punnett square · working sheet

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Record the parent alleles. Fill all four boxes. Then find the chance of aa.

Aa × Aa · side / top	**A**	**a**
A	_____	_____
a	_____	_____

Compare with Aa × aa.

Aa × aa · side / top	**A**	**a**
a	_____	_____
a	_____	_____

Aa × Aa chance of aa: ____ . Aa × aa chance of aa: ____ .

Made from the web lesson SCI-LAUNCH-A2-W05-L2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale